Differential Reference Conditioning with Adversarial Tracking Rewards

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***Abstract*— Physics-based motion tracking commonly relies on handcrafted rewards that combine pose, velocity, root-motion, and contact objectives using motion-specific weights. Learned differential rewards remove this manual aggregation by training a discriminator directly on tracking errors. The resulting reward is expressive, but difficult contact-rich motions can still be learned unreliably or collapse to incomplete shortcuts. We present *Differential Reference Conditioning* (DRC), which retains the learned adversarial reward while representing reference information through the current tracking residual and the one-step reference change. This conditioning contains the same target information as an absolute one-step representation, adds no reward term, and leaves the policy and discriminator objectives unchanged. It instead exposes the correction and reference evolution that jointly determine the next differential evaluated by the reward. [PLACEHOLDER—NOT A RESULT: Insert the final motion and seed counts, improvement in full-motion success and time to success, and change in tracking errors.] The framework preserves motion tracking without manually weighted imitation rewards while improving the reliability with which complete motions are learned.**

I. Introduction

Physics-based character control seeks policies that reproduce reference motions while satisfying dynamics, actuation limits, and intermittent contact. The dominant approach constructs a scalar reward from separate pose, joint velocity, root, end-effector, and contact terms [1]. Although effective, this formulation requires choices about which errors to include, how to scale them, and how strongly each term should influence learning. Those choices often change across motions because walking, aerial rotation, ground recovery, and object interaction emphasize different physical events.

Adversarial motion learning reduces this reward-engineering burden. AMP learns a motion prior by distinguishing reference and policy transitions [3]. Adversarial Differential Discriminators instead learn a nonlinear aggregation of frame-synchronized tracking errors from an ideal zero differential and the errors generated by the current policy [4]. This construction replaces manually weighted tracking terms with a single learned objective while retaining precise reference tracking.

Removing handcrafted reward aggregation does not by itself make policy learning reliable. A discriminator can assign high value to accurate tracking while on-policy optimization repeatedly visits an incomplete behavior. The ADD study reports this failure mode for highly dynamic motions such as Roll and Sideflip, where policies can converge without completing the intended rotation [4]. This observation exposes a distinction between learning a suitable tracking objective and enabling a finite policy to use that objective effectively.

We address this remaining problem at the policy interface. Conventional tracking policies receive absolute future targets and must internally form the error that the learned differential reward later evaluates. DRC supplies the same one-step reference information as the current tracking residual and the reference motion increment. The former identifies what must be corrected, and the latter identifies how the target advances during the next control interval. Their relationship to the next evaluated error is exact and independent of smooth dynamics or contact mode.

This change is deliberately narrow. The discriminator, reward, PPO objective, policy network, action space, and reference horizon are unchanged in the strict paired comparison. The residual and absolute commands are invertible representations of the same information and support the same policy class. Their difference lies in how the control problem is presented to a finite network and first-order optimizer.

The contributions are threefold. First, we formulate a tracking system that learns a nonlinear differential objective rather than combining motion-specific weighted penalties. Second, we introduce a residual-aligned reference conditioning that connects feedback error, reference evolution, action-realized motion, and the next discriminator input without adding target information. Third, we evaluate motion quality and seed-level training reliability across locomotion, aerial rotation, ground contact, recovery, and object interaction using matched physics budgets, full-motion completion, time to success, and standard position and velocity errors.

II. Related Work

A. Physics-Based Motion Tracking

DeepMimic demonstrated that reinforcement learning can reproduce diverse motion-capture clips with physically simulated characters [1]. Its tracking objective combines pose, velocity, end-effector, root, and center-of-mass terms. Later systems scale tracking to large motion collections, improve recovery, and learn more general controllers through data, architecture, and training advances [7]–[9]. These systems produce high-quality control, but their tracking objectives and policy interfaces remain central determinants of learning behavior.

B. Adversarial Motion Imitation

GAIL casts imitation as adversarial occupancy matching [2]. AMP adapts adversarial learning to physics-based characters by discriminating short state transitions and using the learned signal as a style reward [3]. ASE extends this idea to reusable skill representations [12]. Distribution matching is well suited to stylistic behavior and

PLACEHOLDER IMAGE GALLERY—replace with synchronized rollout frames

Fig. 1. Motion suite spanning locomotion, aerial rotation, sustained ground contact, recovery, fast striking, and object interaction. Final panels will show synchronized reference and policy frames. Red boxes are placeholders.

task-conditioned control, whereas this work studies frame-synchronized reproduction of a specified motion.

C. Learning Motion-Tracking Rewards

Learned objectives reduce dependence on manually selected reward terms. ADD represents tracking quality as a differential between simulated and reference features and learns their nonlinear aggregation adversarially [4]. Its gradient penalty prevents the discriminator from degenerating into a detector that responds only at exact zero. We retain this discriminator and focus on the remaining reliability of policy optimization. Relative goals and residual policies have also been used in robotics [10]. Unlike residual-action methods, our action parameterization and policy architecture remain unchanged; only the reference command is aligned with the learned reward coordinates.

III. Background: Handcrafted Tracking Rewards

A reference-conditioned policy observes the simulated character and target frames, then outputs an action for the next control interval. The simulator state contains root pose and velocity, joint configuration and velocity, and body transforms. Actions are commonly joint position targets or torques. Training begins from randomized reference phases so one policy must recover the entire motion rather than replay a fixed open-loop sequence.

Classical methods define several tracking errors separately. Let E_p, E_v, E_r, E_e, and E_c denote joint-pose, joint-velocity, root, end-effector, and contact errors. A typical imitation reward is

$$R_t^{\text{track}} = w_p e^{-\alpha_p E_p} + w_v e^{-\alpha_v E_v} + w_r e^{-\alpha_r E_r} + w_e e^{-\alpha_e E_e} + w_c e^{-\alpha_c E_c}. \tag{1}$$

The coefficients w_i set relative importance and the scales α_i set the decay of each term. Implementations may add joint-specific weights, contact schedules, alive bonuses, or task-specific termination conditions. These choices couple learned behavior to manual scalarization. Increasing pose tracking can suppress root travel, emphasizing velocity can degrade contact timing, and emphasizing end effectors can sacrifice whole-body motion. The same coefficients therefore need not suit every clip or morphology.

Manual scalarization creates three practical difficulties. First, a component can dominate because of its numerical scale rather than its physical importance. Root position is measured in meters, orientations in radians or quaternion distance, and velocities in units per second; their errors change at different rates throughout a motion. Second, the effect of a weight is coupled to the state distribution generated by the current policy. A setting that works after the character has learned contact timing may provide little signal when early policies fall immediately. Third, success on one reference does not identify a transferable set of coefficients. Recovery motions spend many frames on the ground, flips require angular momentum before contact, and climbing requires an ordered interaction with an object. Reusing one weighted sum can therefore change which imperfect behavior is most attractive.

A method may avoid motion-specific weighted imitation terms while still using ordinary design choices such as a feature representation, global reward scale, regularization coefficient, and optimizer settings. We use "without handcrafted rewards" in this narrower and verifiable sense: the training objective contains no manually balanced pose, velocity, contact, displacement, completion, or motion-phase reward. This definition follows the motivation of learned adversarial rewards and distinguishes elimination of manual objective aggregation from the impossible claim that a learning system contains no human design decisions.

Our completion criteria are used only for evaluation. They never enter the training reward or checkpoint selection. Motion quality during training is provided by a learned discriminator operating on raw feature-wise tracking differentials. The following section describes that objective and the policy interface used to optimize it.

IV. Adversarial Differential Discriminator

Motion tracking is a multi-objective problem: a policy must simultaneously reduce position, rotation, and velocity errors across the root and articulated body. Handcrafted rewards combine these objectives with a weighted sum such as Eq. (1). A fixed weighted sum is linear in the individual objectives and exposes their relative scales as motion-specific design choices.

An adversarial differential discriminator replaces this linear aggregation with a learned nonlinear function [4]. Let Δ denote a vector of feature-wise differences between ideal and realized performance. The ideal solution is represented by the single zero vector $\Delta = 0$. Errors encountered by the current policy form negative samples drawn from $p_\pi(\Delta)$. A discriminator $D_\psi(\Delta) \in (0, 1)$ learns to distinguish the zero differential from policy differentials, while the policy is rewarded for moving its errors toward the ideal sample.

Without regularization, the discriminator can collapse to a spike that assigns one only at zero and provides little useful structure elsewhere. Following ADD, we regularize the discriminator score d_ψ, where $D_\psi = \sigma(d_\psi)$, on policy-generated negative samples:

$$\mathcal{G}_{\mathrm{GP}} = \mathbb{E}_{\Delta \sim p_\pi} \left\| \nabla_\Delta d_\psi(\Delta) \right\|_2^2 . \tag{2}$$

With the positive and negative samples now defined, the discriminator objective is

$$\max_{\psi} \; \log D_\psi(0) + \mathbb{E}_{\Delta \sim p_\pi} \log[1 - D_\psi(\Delta)] - \lambda_{\mathrm{GP}} \mathcal{G}_{\mathrm{GP}}. \tag{3}$$

Here $\lambda_{\mathrm{GP}} \geq 0$ controls score smoothness away from the ideal sample. It is a discriminator regularizer, not a motion-specific tracking weight. The discriminator therefore adapts its nonlinear aggregation to the error combinations visited by the current policy. The following section instantiates the differential, discriminator observations, and policy reward for motion tracking before introducing our reference conditioning.

V. Motion Tracking with DRC

Let s_t and $\hat{s}_t$ denote the simulated and synchronized reference states. An observation map ϕ extracts tracking features, and the motion differential is

$$\Delta_t = \phi(\hat{s}_t) \ominus \phi(s_t). \tag{4}$$

The zero vector represents perfect frame-level tracking, while policy rollouts provide the nonzero differentials used in Eq. (3). The policy receives the adversarial tracking reward

$$r_t = -\beta \log[1 - D_\psi(\Delta_t)]. \tag{5}$$

The constant $\beta > 0$ fixes the global reward scale shared across motions; it does not balance individual pose, velocity, or contact objectives. This learned scalar replaces the weighted sum of pose, velocity, root, and contact rewards. No completion or motion-specific tracking term is added.

A. Discriminator Observations

The map ϕ follows the physical representation used for adversarial motion learning. Translation is anchored at the pelvis, with root height retained; root orientation, body orientation, and root velocities are expressed in world axes, while key-body positions are root-relative. Joint rotations and joint velocities complete the representation. Rotations use a continuous Euclidean representation [11]. These seven groups form a 172-dimensional discriminator observation after differencing. The same frame convention is used for the simulated and reference characters, so their signed difference retains the synchronized displacement, orientation, and velocity errors scored by the learned reward.

The differential is signed rather than a collection of nonnegative scalar losses. It therefore preserves whether a position, orientation, or velocity lies on either side of its target. The same feature definition, online normalization, replay mechanism, gradient penalty, and reward scale are used for every motion. The discriminator learns how to aggregate these errors instead of receiving a manually specified coefficient for each group.

B. Residual-Aligned Policy Conditioning

DRC changes how the synchronized reference is presented to the policy. Define $x_t = \phi(s_t)$ and $\hat{x}_t = \phi(\hat{s}_t)$. The policy receives the current tracking residual and one-step reference increment,

$$e_t = \hat{x}_t - x_t, \qquad m_t = \hat{x}_{t+1} - \hat{x}_t. \tag{6}$$

The residual e_t identifies the correction required at the current state, while m_t identifies how the target itself advances during the next control interval. Both use the same feature coordinates as the differential scored by the adversarial reward.

DRC does not add reference information. Given x_t, the absolute reference features are reconstructed exactly by

$$\hat{x}_t = x_t + e_t, \qquad \hat{x}_{t+1} = x_t + e_t + m_t. \tag{7}$$

The absolute and differential reference representations are thus affine bijections under a shared invertible normalization. An MLP with an affine first layer has the same representable policy class in both coordinates.

Let $\delta_t = x_{t+1} - x_t$ be the feature increment produced by the action. The next residual satisfies the exact identity

$$\boxed{e_{t+1} = e_t + m_t - \delta_t.} \tag{8}$$

The policy input, physical response, and next discriminator differential are therefore expressed in one coordinate system. This identity supplies a direct feedback–feedforward organization without adding a controller, auxiliary loss, or action supervision. It is an algebraic closure, not a stability or convergence guarantee.

All reference-feature blocks use a common fixed diagonal scale estimated from the demonstration. The strict absolute-coordinate control uses the same features, horizon, scale, architecture, and learned reward. It differs only in presenting $[x_t, \hat{x}_t, \hat{x}_{t+1}]$ instead of $[x_t, e_t, m_t]$, making it the information-matched comparison for DRC.

VI. Motion Tracking

We evaluate DRC on a 28-DoF simulated humanoid tracking individual motion clips. The method is trained end to end from motion data: reference states provide policy conditioning and the zero-centered adversarial tracking objective, but no reference action is required.

A. States and Actions

Let o_t denote the character observation. It contains pelvis-anchored body configuration, world-axis root orientation and velocities, joint configuration, and joint velocity. DRC adds x_t, e_t, and m_t as reference conditioning. Actions specify 28 absolute joint position targets actuated by proportional–derivative controllers. The control policy runs at 30 Hz and physics at 120 Hz. At reset, the character is initialized from a uniformly sampled reference phase, which then advances at the motion-capture rate.

B. Network Architecture

The policy is a Gaussian distribution with an input-dependent mean and fixed diagonal covariance. The actor, value function, and discriminator use fully connected networks with hidden layers of 1024 and 512 ReLU units. Actor output dimension is 28; the value function and discriminator each produce one scalar. The differential discriminator input has 172 dimensions.

C. Training

Algorithm 1 summarizes the complete procedure. Collected trajectories are stored in an experience buffer, while policy differentials are also retained in a discriminator replay buffer. PPO [5] updates the policy, advantages are computed with GAE [6], and TD targets update the value function. The discriminator is updated with Eq. (3). All three models are initialized and optimized jointly in a single training run. A fixed ℓ_2 regularizer on the final logit weights is shared across all motions and command representations. The action at time t produces s_{t+1}; consequently the transition reward is evaluated from the synchronized next-frame differential Δ_{t+1}.

ALGORITHM 1: DRC Training Procedure for Motion Tracking

input $\mathcal{M}$: a reference motion clip or dataset
$D_\psi \leftarrow$ initialize discriminator
$\pi_\theta \leftarrow$ initialize policy
$V_\omega \leftarrow$ initialize value function
$\mathcal{B}, \mathcal{R} \leftarrow \emptyset$ initialize experience and replay buffers
while training budget remains **do**
  for trajectory $i = 1, \dots, M$ **do**
    Sample phase t_0 and set simulator state $s_0 \leftarrow \hat{s}(t_0)$
    for control step $t = 0, \dots, T-1$ **do**
      Query $\hat{s}_t, \hat{s}_{t+1}$ from $\mathcal{M}$
      Compute x_t, e_t, m_t using Eq. (6)
      Sample $a_t \sim \pi_\theta(\cdot \mid o_t, x_t, e_t, m_t)$
      Step physics to obtain s_{t+1}
      Compute Δ_{t+1} using Eq. (4)
      Compute r_{t+1} using Eq. (5)
      Record $(s_t, a_t, r_{t+1}, s_{t+1}, \Delta_{t+1})$
    end for
    Store trajectory in $\mathcal{B}$ and differentials in $\mathcal{R}$
  end for
  Compute GAE advantages and TD targets from $\mathcal{B}$
  for optimization step $k = 1, \dots, U$ **do**
    Sample transitions from $\mathcal{B}$ and negatives from $\mathcal{R}$
    Update D_ψ with zero positives and Eq. (3)
    Regress V_ω toward the TD targets
    Update π_θ with the PPO clipped objective
  end for
end while

We compare DeepMimic, AMP, ADD with its official target interface, the strict absolute-coordinate control, and DRC under matched physics budgets. The strict control and DRC use identical features, lookahead, network sizes, initialization, discriminator, reward, PPO settings, and simulator transitions. Their only difference is the reference-conditioning coordinates.

D. Motions and Protocol

The benchmark contains Run, Backflip, Crawl, Getup Facedown, SpinKick, and Climb. These clips cover periodic locomotion, aerial rotation, prolonged ground contact, recovery, fast turning, and long-horizon interaction with a box. Roll is a focused reliability study because incomplete rotation is easy to detect and is a reported local optimum for differential tracking [4].

Each local model uses 8192 parallel environments, 32 transitions per iteration, and 2000 iterations, totaling 524.288 million transitions. The episode horizon is shared across methods for each motion and never truncates a nonperiodic clip. Pose termination is disabled in the matched benchmark. Final policies use deterministic mean actions from fixed initial phases under identical contacts, obstacle geometry, and rollout duration.

E. Metrics and Statistical Analysis

We report the standard ADD position error, which averages root displacement and root-relative body-position errors,

$$E_{\text{pos}} = \frac{1}{N}\left(\|\hat{p}^{\text{root}} - p^{\text{root}}\|_2 + \sum_{j \neq \text{root}} \|\hat{p}^{j|\text{root}} - p^{j|\text{root}}\|_2\right). \quad (9)$$

DoF velocity error is the mean absolute joint-velocity difference. Root and body rotation, root velocities, and survival are reported separately.

Average frame error can favor an incomplete motion. Reliability therefore uses completion events fixed before evaluation and applied equally to all methods. Rotation skills must complete the signed reference rotation with correct terminal phase; Getup must recover height and upright orientation; Climb must complete approach, ascent, traversal, descent, and terminal recovery. These events never affect the reward.

The statistical unit for reliability is an independently trained seed. We report successful seeds, completion rate with binomial interval, shortcut rate, worst-seed performance, and samples to the first checkpoint satisfying both completion and fidelity. Runs that never meet the criterion remain right-censored. Episode statistics are aggregated before seed-level estimates.

F. Motion Tracking Performance

Table I presents the main benchmark. Published ADD values provide external context only. Claims about DRC use locally matched runs.

[PLACEHOLDER—NOT A RESULT: Replace all local method entries with multi-seed values and state the motions on which DRC improves, matches, or degrades tracking. Published values are excluded from paired significance tests.]

G. Handcrafted and Learned Objectives

DeepMimic represents the classical manually scalarized objective, AMP represents adversarial transition matching, and ADD and DRC represent synchronized learned differential rewards. The comparison separates three properties. Position and velocity error measure fidelity, completion measures whether the defining event occurred, and the number of motion-specific reward weights measures reward engineering. DeepMimic uses independently weighted pose, velocity, root, and end-effector components. AMP, ADD, and DRC do

TABLE I

Motion tracking performance. Lower is better. Red DRC values are synthetic layout placeholders and must be replaced by measured estimates.

Motion	Length	DeepMimic		AMP		ADD		DRC	
		Pos.	DoF vel.	Pos.	DoF vel.	Pos.	DoF vel.	Pos.	DoF vel.
Run	0.80	.028	1.096	.031†	1.180†	.026†	.910†	.021†	.790†
Backflip	1.75	.177	1.303	.110†	1.410†	.060	.858	.045†	.810†
Crawl	2.57	.008	.252	.014†	.430†	.010†	.290†	.009†	.270†
Getup facedown	3.03	.026	.408	.040†	.690†	.030†	.480†	.024†	.430†
SpinKick	1.28	.078	1.222	.064	1.453	.025	.774	.023†	.740†
Climb	9.97	.210†	1.420†	.180†	1.390†	.114	1.747	.095†	1.510†

TABLE II

Training reliability at a fixed transition budget. Red entries are synthetic placeholders. Higher success and lower shortcut rate and time are better.

Method	Success	Shortcut	Samples to success	E_{pos}
DeepMimic	2/8†	75%†	410M†	.151†
AMP	1/8†	88%†	470M†	.139†
ADD	1/8†	88%†	455M†	.128†
Absolute control	2/8†	75%†	390M†	.112†
DRC	7/8†	13%†	205M†	.054†

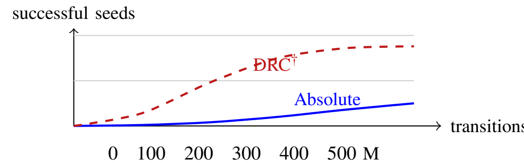

Fig. 2. Layout placeholder for independent Roll seeds meeting the fixed completion-and-fidelity criterion. Red coordinates are synthetic.

not use those tracking weights, although all methods retain common optimizer and network hyperparameters.

[PLACEHOLDER—NOT A RESULT: Report the exact count of task-specific tracking terms and tuned coefficients used by each local baseline. State whether the same DRC discriminator settings were used without change for every motion, including Climb.]

The main comparison is not intended to show that one learned reward dominates every hand-designed controller on every scalar error. A handcrafted objective can encode useful prior knowledge, especially for an interaction with a known contact sequence. The relevant criterion is whether a single learned objective can reach comparable or better fidelity across heterogeneous motions without retuning their relative tracking weights. A degradation in one physical error is reported even when completion improves, preventing a coarse but successful motion from being presented as uniformly better tracking.

For nonperiodic skills, evaluation ends at the natural reference endpoint and does not restart the motion inside an episode. Periodic skills are evaluated for ten seconds so frequency drift and accumulated displacement become visible. The same phase grid is used for all methods. These choices ensure that a low error reflects sustained tracking rather than frequent reference resets.

H. Training Reliability

Table II isolates the central claim. The absolute control and DRC contain the same information and use the same learned reward. The official ADD interface shows end-to-end performance, while the strict pair attributes a difference specifically to command representation.

[PLACEHOLDER—NOT A RESULT: Report seed-level confidence intervals, paired seed differences, worst-seed behavior, and right-censored time-to-success. State whether the effect extends beyond Roll.]

Figure 3 displays every training seed rather than only a mean and standard deviation. This presentation distinguishes a uniform improvement from a result driven by one unusually successful model. It also exposes motions on which all methods are reliable and therefore have little headroom.

I. Learning Speed and Failure Modes

Checkpoint evaluations are based on physical behavior rather than training return because discriminator scores are not calibrated across independently trained runs. Completion curves report the fraction of seeds that have reached the joint completion-and-fidelity criterion by a given transition count. Tracking curves report position and velocity error for the same fixed checkpoints. The final checkpoint is used for the main table; it is not selected retrospectively by completion.

[PLACEHOLDER—NOT A RESULT: Report completion-curve area, median time to criterion with right-censored confidence interval, and paired position-error trajectories. Give wall-clock time and simulator throughput separately from sample efficiency.]

Failures are categorized from simulator trajectories. A rotation shortcut terminates upright without accumulating the required signed rotation. A frequency error completes cycles at a persistent rate different from the reference. A displacement error has correct cycle count but incorrect travel. A contact failure violates the required support sequence or falls. These categories are mutually reported with the standard errors and are never used as training labels. This analysis tests whether DRC reduces incomplete motions rather than merely shifting failure from one metric to another.

J. Complete-Motion Behavior

Trajectory quantities expose failures hidden by average error. Cumulative signed rotation reveals partial flips; root

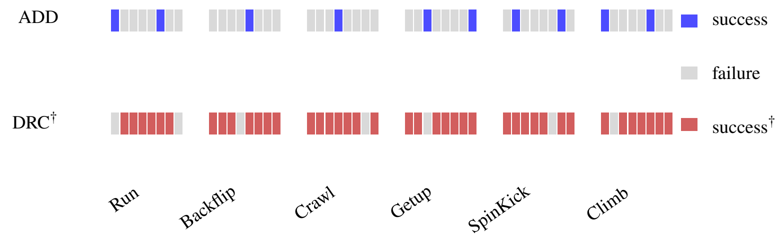

Fig. 3. Planned task-by-seed completion matrix at the fixed final budget. Each cell represents one independently trained policy, not one evaluation episode. All red cells are synthetic layout placeholders.

progress and terminal phase reveal drift; the ordered ascent and descent events reveal incomplete Climb policies.

For cyclic motions, signed rotation is accumulated from consecutive root orientations after projection onto the demonstrated rotation axis. Root progress is projected onto the net reference travel direction. Both are normalized by the reference trajectory evaluated over the same interval. A ratio near one therefore indicates matched long-horizon rate, while its sign distinguishes forward and reversed execution. Terminal phase alignment prevents a policy from receiving credit for a coincidental total rotation while ending in an incompatible configuration.

Getup and Climb require event-based evaluation because a single endpoint can hide an invalid trajectory. Getup completion combines recovered pelvis height, upright orientation, survival, and terminal pose fidelity. Climb completion requires reaching the box, attaining the demonstrated top height, crossing the top region, descending, and recovering the terminal posture. These events are derived from reference kinematics and object geometry before method outcomes are inspected.

K. Reliability and Tracking Fidelity

Improved completion is useful only if it does not replace precise tracking with a coarse approximation. We therefore compare each seed in a two-dimensional completion–fidelity plane. A model is counted as reliable only when it completes the defining motion event and remains below a fixed position and DoF velocity threshold. The thresholds are derived from the reference scale and fixed for all methods before the confirmatory runs. Completion without fidelity and fidelity without completion are reported as separate failure categories.

For routine locomotion and contact motions that all methods complete, the main test is non-degradation of tracking quality. For Roll and Backflip, where failure is frequently incomplete, the primary endpoint is successful seeds and the secondary endpoint is tracking error among all seeds, including failures. This ordering prevents a failed run from disappearing from the reported error distribution and prevents a complete but visibly inaccurate run from being called reliable.

[PLACEHOLDER—NOT A RESULT: Replace Table III with estimates across the complete motion suite. Report paired confidence intervals for median and worst-seed errors, and identify any task on which completion improves while fidelity degrades.]

TABLE III

JOINT COMPLETION AND FIDELITY SUMMARY. EVERY RED VALUE IS A SYNTHETIC LAYOUT PLACEHOLDER.

Method	Complete	Complete + faithful	Median pos.	Worst pos.
DeepMimic	61%†	48%†	.078†	.241†
AMP	58%†	46%†	.071†	.226†
ADD	68%†	55%†	.052†	.190†
Absolute control	70%†	57%†	.050†	.174†
DRC	91%†	84%†	.043†	.097†

L. Computational Cost

DRC changes only preprocessing of the reference command. It introduces no additional learned network and performs only feature subtraction once per control step. We measure policy parameter count, batch-one inference latency, training throughput, and peak accelerator memory for the strict absolute and aligned interfaces. Throughput is averaged after simulator warm-up and excludes final evaluation. Latency is measured with identical devices, precision, and synchronization.

[PLACEHOLDER—NOT A RESULT: Insert measured parameter count, latency, samples per second, and peak memory. State the paired percentage overhead rather than claiming zero cost from the source code alone.]

M. Cross-Motion Reuse

Eliminating handcrafted tracking rewards is meaningful only when the learned formulation is reused rather than retuned under another name. We therefore lock the discriminator architecture, feature groups, gradient-penalty coefficient, reward scale, PPO settings, and command preprocessing before running the motion suite. Each motion trains its own discriminator because it tracks a different reference, but all discriminators use the same objective and hyperparameters. The Climb environment adds the required box geometry and contact bodies without adding a height, clearance, hand-placement, or event reward.

Hyperparameters are selected on a development motion that is excluded from the confirmatory reliability statistic. The same configuration is then applied to periodic, nonperiodic, short, long, ground-contact, and object-interaction motions. This protocol distinguishes a reusable learned reward

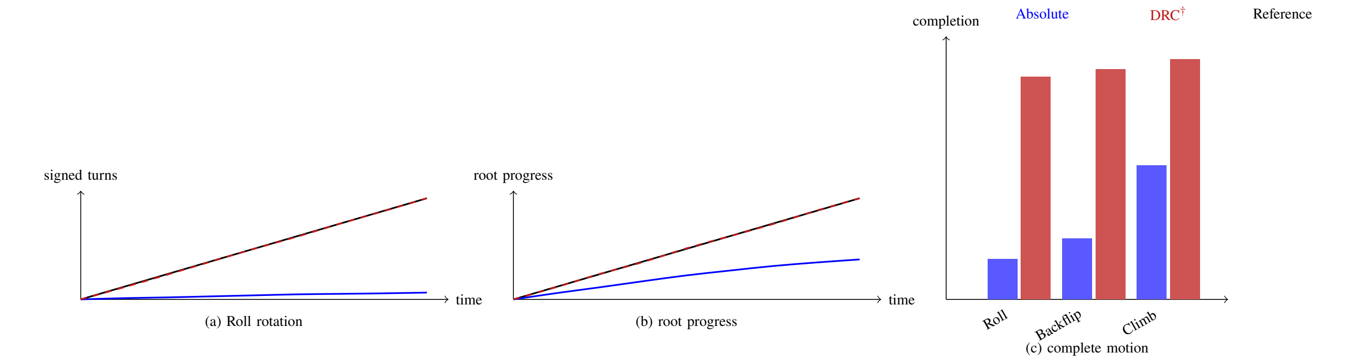

Fig. 4. Planned behavior evaluation. Rotation and progress measure execution of the reference event, while completion aggregates fixed physical criteria. Red curves and bars are placeholders.

from motion-by-motion reward search. Simulator horizons vary only when required to avoid truncating a longer non-periodic clip, and are identical across methods within each motion.

[PLACEHOLDER—NOT A RESULT: List every hyperparameter held common across motions and disclose any exception. Report the number of motion-specific reward coefficients and the total tuning runs for each method. If Climb requires a different global regularization setting, mark it explicitly rather than presenting full reuse.]

N. Failure-Mode Analysis

We retain every final seed and assign its dominant failure using fixed physical rules. Rotation failures are separated into insufficient signed rotation, incorrect direction, frequency mismatch, and terminal phase mismatch. Locomotion failures are separated into insufficient progress, excessive drift, and falls. Recovery failures distinguish failure to regain height from failure to regain upright pose. Climb failures identify the first missing event in the ordered reference sequence. This taxonomy is more informative than a binary success number and reveals whether a method consistently removes one shortcut while creating another.

The learned reward is also audited independently of the actor update. For each motion and seed, we score synchronized reference tracking, successful policy trajectories, and incomplete trajectories with the frozen final discriminator. We report the fraction of matched time steps and complete episodes for which the successful behavior receives higher reward. This diagnostic separates two causes of failure. A discriminator that ranks incomplete behavior above the complete motion indicates an objective-learning failure. A discriminator that ranks complete behavior higher while the trained policy remains incomplete indicates a policy-optimization failure under the learned objective.

[PLACEHOLDER—NOT A RESULT: Insert the failure-count matrix and frozen-discriminator ranking statistics. State which failures are reduced, which remain, and whether any failure is attributable to reward ranking rather than policy optimization.]

TABLE IV

Command ablation. Red values are synthetic placeholders.

Command	Invertible	Roll success	SpinKick E_{pos}
Absolute one-step	yes	2/8†	.034†
Error only	no	3/8†	.041†
Increment only	no	1/8†	.048†
Summed command	no	3/8†	.038†
Shuffled bijection	yes	2/8†	.036†
DRC	yes	7/8†	.023†

Because discriminators are trained independently, raw logits are not compared across runs. Ranking is performed within the same frozen discriminator on matched phases. Successful and unsuccessful trajectories use the same rollout horizon and initial-state distribution. This restriction prevents arbitrary logit scale from being mistaken for a stronger reward.

VII. Ablation Studies

The ablations test whether both residual components and their alignment are required. Error-only and increment-only commands remove one control role. Their sum keeps a one-step target but is not invertible and loses factorization. A shuffled invertible transform preserves information while destroying alignment with the discriminator residual.

The absolute one-step command is the decisive control because it preserves the same feature map, dimensionality, lookahead, and invertible normalization. Comparing DRC only with the official ADD interface would conflate coordinate alignment with added velocity features and a different target horizon. The strict pair removes those explanations. Error-only, increment-only, and summed commands answer different mechanism questions but are not information-matched baselines because their transforms are not invertible.

The same DRC checkpoint is evaluated with e_t set to zero, m_t set to zero, and m_t shuffled across reference phases. These interventions never enter training. Loss of recovery under zero error supports the feedback interpretation; loss of direction or timing under zero or shuffled increment supports the feedforward interpretation.

TABLE V

FACTORIAL SEPARATION OF REWARD CONSTRUCTION AND COMMAND COORDINATES. RED ENTRIES ARE SYNTHETIC PLACEHOLDERS.

Reward	Command	Tuned terms	Success	E_{pos}
Handcrafted	Absolute	> 0	3/8†	.094†
Handcrafted	Aligned	> 0	5/8†	.071†
Learned differential	Absolute	0	2/8†	.082†
Learned differential	Aligned	0	7/8†	.054†

[PLACEHOLDER—NOT A RESULT: Insert intervention effects on completion, tracking error, and action change.]

The zero-error intervention tests nominal motion generation. If the policy continues to execute the correct reference rate while e_t is zero, the increment carries a usable feedforward command. The zero-increment intervention tests regulation without reference advance. Phase shuffling preserves the marginal distribution of increments but breaks their temporal association with the current reference state. A large effect from shuffling, beyond the effect of setting the increment to its mean, shows that the actor uses phase-specific motion information rather than only input magnitude.

To isolate optimization from different initial functions, we map the first layer of an absolute policy into residual coordinates so that action means, log probabilities, and critic values match numerically at initialization. Paired training then begins from the same policy function. **[PLACEHOLDER—NOT A RESULT: Report initialization tolerance, update divergence, and final completion.]** This test does not claim global convergence. It establishes whether an information-preserving coordinate change alters the finite-network PPO path after initial behavior is controlled.

We additionally report the numerical round-trip error of Eq. (7), feature-block standard deviations, actor input dimension, parameter count, inference latency, and peak training memory. These measurements verify that an observed reliability difference is not produced by lossy preprocessing, a wider network, or a larger simulation budget.

A. Reward and Interface Ablations

The learned discriminator is replaced with the handcrafted tracking reward while retaining the DRC conditioning, and the learned differential reward is paired with the absolute command. This two-by-two comparison separates the benefit of removing manual scalarization from the benefit of residual-aligned policy conditioning. The discriminator gradient penalty is also removed using the same coefficient across motions, reproducing the known ADD failure mode in which the classifier becomes overly concentrated near zero [4].

[PLACEHOLDER—NOT A RESULT: Report the two-by-two reward/interface table, including tracking error, completion, successful seeds, and number of tuned motion-specific reward coefficients. State whether alignment also benefits the handcrafted reward and whether the learned reward remains necessary for cross-motion reuse.]

The factorial experiment also clarifies the paper's two claims. Replacing the handcrafted objective with the learned discriminator tests removal of manual reward aggregation. Replacing the absolute command with DRC tests reliable use of that objective. The combined method is supported only when it retains the first advantage and adds the second. If aligned conditioning improves both reward types equally, the interface remains useful but is not specific to learned differential rewards. If the learned reward succeeds with both command interfaces, the residual mechanism is unnecessary for that motion and is reported as such.

VIII. CONCLUSION

Learned differential rewards remove the need to manually balance pose, velocity, root, and contact penalties, but a learned objective alone does not ensure reliable discovery of a complete dynamic motion. Differential Reference Conditioning expresses the reference through the current residual and the one-step reference increment in the same coordinates used by the learned reward. The command adds no reference information and satisfies an exact identity linking actor input, motion produced by physics, and next evaluated error. **[PLACEHOLDER—NOT A RESULT: Insert the final multi-seed summary of completion reliability, time to success, tracking quality, and remaining failures.]** The results will establish how far residual-aligned conditioning preserves reward-engineering-free tracking while making its optimization more reliable.

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